

Mostafa Mashkani (Python Developer)

Contact Information

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About Me

Senior Python Developer with over **9 years of hands-on experience** designing and scaling robust serverside systems. Expertise in **Django**, **Django Rest Framework (DRF)**, and **FastAPI**, with a strong focus on performance, reliability, and clean architecture. Successfully led an AI engineering team, driving the development of intelligent services and high-throughput APIs. Passionate about building maintainable codebases and delivering backend solutions that support complex product needs

Work Experience

Python Backend Developer - Advanced Analytics Australia

April 2025 - Present

- Designed and implemented high-performance **RESTful APIs using Django and Django REST Framework (DRF)** for real-time detection, tracking, and identification of individuals in large-scale surveillance environments, with automated data persistence pipelines.
- Improved large-scale image search performance by 60% through **Apache Solr** optimization, enabling efficient indexing and retrieval across datasets containing 1M+ images.
- Developed full-stack web applications using Django and React.js, with emphasis on secure API design, authentication workflows, and production-grade backend architecture.
- Developed industrial safety monitoring systems using **Django REST Framework (DRF)** to automate safety-control workflows in factory environments, integrating **WebRTC streaming pipelines** for real-time Full HD video transmission between backend services and frontend clients.
- Built scalable backend infrastructure with **JWT authentication**, **Redis-based caching**, and optimized **Django ORM** database interactions, significantly improving response times and reducing repeated database load.
- Designed and implemented **real-time event streaming pipelines using Apache Kafka**, replacing traditional FIFO job queue architecture to improve throughput, asynchronous processing reliability, and service decoupling across distributed backend components.
- Engineered a **custom health-checking and observability framework** for camera devices and backend services, enabling automated service monitoring, fault detection, and proactive recovery across distributed

infrastructure.

- Managed and optimized real-time processing pipelines for 6 simultaneous HD camera streams, achieving **end-to-end latency below 80ms** for detection and tracking workloads under production conditions.
- Data-streaming solutions using Django and gRPC for real-time synchronization between edgeprocessing devices and backend.
- Integrated **Prometheus-based observability** and monitoring systems, enabling real-time metrics collection, backend performance analysis, service health monitoring, and operational visibility across Django services and distributed microservice architecture.

Python Backend Developer - ITSAAZ

February 2022 - November 2024

- **Designed and deployed a scalable microservices architecture** using **FastAPI** for secure authentication system, including face recognition, video-based sentence similarity checks, and OCR processing of national ID cards.
- Built **Django-based** systems for person detection on CCTV streams, including **real-time video transmission** over UDP socket, recognition, and traffic counting, leveraging **PostgreSQL** and **Apache Solr** with feature-vector clustering to handle high-volume data workloads.
- Designed and optimized **RNN models** for human-behavior recognition and activity analysis.
- Containerized core services with **Docker** and created end-to-end **CI/CD pipelines** to automate deployment, testing, and ensure system reliability.

Python Developer - QUANTRISK

May 2021 - November 2021

- Built **Django-based** Urban electricity consumption forecasting service.
- Applied **multiprocessing and Pool Processing** for asynchronous forecaster model training.
- Collaborated on the development of scalable web applications using Python and Django.
- Forecasting electrical power using **RNN models** , **XGBoost** , **LightGBM** ,..

Python & C# Developer - FARAND SYSTEM

September 2019 - October 2020

- Applied **computer vision and image-processing** techniques for automated fault detection and quality-control workflows.
- Integrated **NVIDIA Jetson Nano** (Python) with **C# server systems**, implementing video streaming on socket between Python services and Windows Form applications, and managing data synchronization with **SQL Server**.
- Contributed to the development of an **intelligent robotic platform (MOOR III)**, supporting perception, control, and system integration components.

Education

M.Sc. Electrical Telecommunications Engineering - Malek e' ashtar University of Technolog

2017 - 2020

- Specialized in Machine Learning and Deep Learning for AI solutions.
- Advanced knowledge in Machine Vision

B.Sc. Electrical Control Engineering - Sahand University of Technology

2013 - 2017

- Start to learning C# programming and Image processing and opencv
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Skills

Technical Skills

- **Programming Languages:** Python, C#, Golang experience
- **Frameworks:** Django, Django Rest Framework (DRF), FastAPI
- **Databases:** PostgreSQL, MySQL, SQL Server
- **Microservices and APIs:** RESTful APIs, Design Patterns, gRPC
- **DevOps and Tools:** Docker, CI/CD, Linux
- **Queue Systems:** Redis, RabbitMQ, Celery
- **Version Control:** Git

Soft Skills

- **Collaboration:** Strong teamwork across multi-disciplinary teams.
- **Problem-Solving:** Analytical and creative approach to tackling complex challenges.
- **Communication:** Clear and effective communication with stakeholders and team members.
- **Adaptability:** Thrives in dynamic environments with evolving technologies.